

Technology Stack

Date	2
Team ID	
Project Name	Personalized Networking Assistant
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S. No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side (e.g., React, Angular, HTML/CSS)	<i>Streamlit</i>	<i>Used to build a simple, interactive web interface where users enter event descriptions, interests, generate conversation starters, request fact verification, and view results without requiring HTML, CSS, or JavaScript.</i>

2	Backend / Server-Side <i>(e.g., Node.js, Python/Django, Java)</i>	<i>FastAPI (Python)</i>	<i>Provides REST APIs for communication between the frontend and AI modules. Handles user requests, coordinates NLP processing, performs fact verification, and returns responses efficiently.</i>
3	Database / Data Storage <i>(e.g., MySQL, MongoDB, PostgreSQL)</i>	<i>Local JSON (history.json)</i>	<i>Stores generated conversation history, fact verification results, and user feedback locally. A lightweight solution suitable for the current implementation without requiring a database server.</i>
4	Cloud / Hosting / Deployment <i>(e.g., AWS, Azure, Heroku, Docker)</i>	<i>Hugging Face Transformers (Local Models)</i>	<i>Runs DistilBERT for event theme extraction and GPT-2 for conversation starter generation locally, eliminating the need for cloud-hosted inference during development.</i>
5	Version Control & CI/CD	<i>Git & GitHub</i>	<i>Used for source code version control, collaboration among team members, project backup, and maintaining development history.</i>

	<i>(e.g., GitHub, GitLab, Jenkins)</i>		
6	Third-Party APIs / Other Tools <i>(e.g., Stripe, Twilio, Google Maps)</i>	<i>Wikipedia API, Pytest, Python-dotenv</i>	<i>Wikipedia API provides fact verification. Pytest is used for unit testing backend modules. Python-dotenv manages environment variables and project configuration securely.</i>